Retrieval-Augmented Generation for Autonomous SOP Creation: A Methodology for Document Synthesis Across Business Functions

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***Abstract*—This Digital Operations for various business functions like Finance & Accounting, Customer Care, Supply Chain Management and others, performed by human operators are extremely reliant on sequential process steps captured in Standard Operating Procedures (SOPs). However, the creation of SOPs is highly cumbersome, resource-intensive (cost, time & effort), and time consuming. The motivation of this research is to assist human operators with an autonomous approach for creating baseline SOPs by leveraging existing SOPs in enterprise knowledge corpus. This experimental research aimed at developing a generalized methodology that can ingest SOPs in different formats and generate baseline SOPs in pre-defined business function-specific formats. The research yielded real-world scalable business applications of Retrieval Augmented Generations with Integrated Toolset design pattern at optimal cost. For the production-grade operationalization of such applications, the study focused on a structured approach involved in the selection of search algorithm during the retrieval process. The study mentions the performance of different search algorithms like Hierarchical Navigable Small Worlds (HNSW), Exhaustive K-Nearest Neighbors (X-KNN), and Full-Text search based on evaluation parameters like Latency, Cost and Accuracy. These experimental analyses unfolded the document synthesis system's capability to leverage latent process steps hidden in existing SOPs and reduce the cycle time, operational cost and effort involved in creating SOPs.**

Keywords— SOPs, productivity enhancement, time efficiency, document workflow optimization, data extraction, HNSW, xKNN, Agent, Tool, error minimization, autonomous, manual process limitations, corporate efficiency, technological advancement.

I. Introduction

In the modern business landscape, organizations across various industries rely heavily on documentation to streamline their processes, ensure consistency, and maintain quality standards. Standard Operating Procedures (SOPs), invoices, and manuals are among the most common types of documents that require frequent creation and updating. However, the traditional approach to creating these documents often involves manual effort, leading to significant time expenditure and potential inconsistencies. This paper introduces a novel solution to address these challenges: a unique document synthesis mechanism designed to automate the generation of new comprehensive documents from existing ones.

The document generation mechanism is developed by experimenting the document creation process on advanced algorithms and data extraction techniques. It aims to eliminate the need for end agents to start writing SOPs, invoices, or manuals from scratch, thereby saving valuable time and resources.

This mechanism is based on dual approach of using queries for context from existing documents and prompts for instructions to preserve pre-defined formatting of the business.
Upon retrieving the search results from existing documents, the information is passed to a Large Language Model (LLM), which is tasked with refining and structuring the responses based on the prompts provided. This structuring is a critical step, as it transforms raw data into coherent and logically organized content.

A noteworthy aspect of our mechanism is its application of asynchronous operations for the document generation process. This architectural decision significantly reduces the time required to generate the final document, thereby enhancing efficiency.

The significance of this research lies in its potential to transform document-intensive processes across various domains. SOPs play a crucial role in maintaining standardized procedures and ensuring compliance with industry regulations. Invoices are essential for financial transactions and record-keeping. Manuals provide detailed instructions

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and guidelines for products or services. By automating the creation of these documents, organizations can achieve significant time savings, reduce human errors, and improve overall efficiency. Previous research in the field of document automation has focused on various aspects, such as template-based generation, natural language processing, and prompting techniques. However, the document synthesis mechanism proposed in this paper takes a novel approach by leveraging the content and structure of existing base files to generate new documents. This approach ensures that the generated documents maintain consistency with the original structure while allowing for necessary modifications and additions.

The main objectives of this research are as follows:

A. *To develop a robust retrieval strategy for effectively extracting relevant data from existing documents.*

B. *To experiment and benchmark on performance of search algorithms- such as Hierarchical Navigable Small Worlds (HNSW), Exhaustive K-Nearest Neighbors (x-KNN), and Full-Text Search - on Latency, Cost and Accuracy.*

C. *Explore and evaluate the potential applications and benefits of the Retrieval Augmented Generation (RAG) mechanism across various business functions.*

II. Literature Review

This section provides an overview of the existing literature related to document automation, focusing on key techniques, approaches, and their applications in various domains.

Template-based document generation has been a common approach in the field of document automation. Timotej Betina et al. [1] proposed an information retrieval system which was integrated with a text editor to find similar documents. Text from different document formats and source code is extracted, and it is then divided into logical and semantic chunks as part of the suggested retrieval technique. An ideal matching method is then applied to these segments in order to determine the degree of similarity with the new content.

Natural Language Processing (NLP) techniques have also been applied to document generation tasks. Gatt and Krahmer [2] provided a comprehensive review of NLP techniques for text generation. They discussed various approaches, including rule-based methods, machine learning techniques, and deep learning models. The authors highlighted the potential of NLP in coherent text-to-text generation.

Several researchers have focused on specific document types and domains. Ghosh et al. [3] proposed a framework for automated generation of legal contracts. They employed a domain-specific ontology and rule-based techniques to generate contract clauses based on user inputs. The authors demonstrated the potential of their framework in reducing the time and cost associated with contract drafting.

In the context of financial documents, Pal. B et al. [4] presented a system for automated generation of financial reports. They utilized a template-based approach combined with data extraction and natural language generation techniques to produce structured financial reports. The authors highlighted the benefits of their system in terms of efficiency and accuracy compared to manual report generation.

The healthcare domain has also benefited from document automation research. James Hunter et al.[5] developed a system for automated generation of nursing shift summaries. They employed natural language processing techniques to extract relevant information from patient records and generate concise shift summaries. The authors reported improved communication and decision-making among healthcare professionals using their system.

While the aforementioned studies have made significant contributions to the field of document automation, they primarily focus on specific techniques or document types.

III. Methodology

A. *Industry generic formats*

After assessing the use case requirements when it comes to curating extensively detailed documents which will be used by the masses on a regular basis. The retrieval mechanism has to be curated in such a way that the information in the query results is relevant for majority of the applications.

B. *Agentic Architecture*

1) *Agent*

The agent is an LLM instance and acts as the orchestrator, It decides which tools to be used based on the prompt strategy devised in the mechanism.

2) *Tool*

It is a function type which is called only for executing a sole task and will only execute that task. In the implementation each tool corresponds to a query engine on a specific document such that the agent has *n* tools for every *n* documents.

The architecture works as a single agent – multi tool framework where a single agent orchestrates multiple tools providing the LLM comprehensive corpus of the entire base document structure.

C. *Dynamic Tool Creation*

Tools are fabricated on the fly as per the base files for the document synthesis. This gives the mechanism flexibility and adaptiveness when it comes to incorporating several document inputs on a fixed business domain.

D. *Search algorithm selection*

In order to efficiently retrieve the most relevant information from existing corpus, the study lead to conducting a comparative experimental analysis of multiple search algorithms including vector search algorithm like HNSW, (*x*-KNN), and Full-Text search. The goal was to assess each algorithm's performance concerning the accuracy of generated documents and computational efficiency.

1) *Hierarchical Navigable Small World (HNSW)*

HNSW[6] is a graph-based search algorithm that excels in finding approximate nearest neighbors in high-dimensional spaces. Its hierarchical structure allows for faster query times by traversing smaller graphs at higher layers before moving to more dense graphs at lower layers.

2) *Exhaustive Nearest Neighbors (xKNN)*

x-KNN[7] is a traditional algorithm that guarantees the retrieval of the exact nearest neighbors. While it is highly accurate on small scale, its performance deteriorates rapidly

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with the increase in the dimensionality of the data and the size of the dataset.

3) Full-text Search

Full-text[8] search algorithms are commonly used for searching textual data within documents or databases. They are optimized for keyword matching and relevance ranking, but they do not inherently support the complexities of high-dimensional vector space search.

IV. Document Synthesis Mechanism

The mechanism is designed to automate the process of generating documents from selected input documents utilizing search algorithm interfaced with a vector store for data retrieval and a Language Learning Model (LLM) for augmentation. The architecture ensures the contextual correctness and syntactical formatting of the outputs, culminating in a refined document. The following subsections detail each component of the system:

A. User Input Processing

The initial stage of the process begins with the collection of user inputs, which include base documents. These inputs are vital for tailoring the output to the user's specific requirements.[9]

$$\begin{aligned}&\textit{Let } \mathrm{U} \textit{ be the set of all documents corpus,}\\&\textit{Consider, a document } D_1 \in U,\\&\text{which } \textit{consists of individual contextually meaningful}\\&\textit{sections } S_1, S_2, .. S_{\mathrm{n}}\\&\textit{we can define Document } D_1 \in U\\&\textit{as a concatenation of sections using function } f\\&\qquad :.\ f(S_1, S_2, .. S_{\mathrm{n}}) = D_1 \quad (1)\end{aligned}$$

B. Query, Data Retrieval and Search Algorithm

Once the user inputs are gathered, they are utilized to modify pre-fixed queries and prompts. The modification process is twofold: the query is adjusted to ensure the content's contextual relevance,

$$\begin{aligned}&\textit{Query } Q_1 \textit{ is generalised for a Retrieval method } R\\&\textit{such that } R \textit{ retrieves from a corpus of } N \textit{ documents,}\\&\qquad R(Q_1) = [S_{1-D_1}, S_{1-D_2}, .. S_{1-D_N}] \quad (2)\end{aligned}$$

The search algorithm(retrieval method) is a core component of the system, responsible for processing the formulated query. It operates in conjunction with a text/vector store, which is a structured repository of indexed data. The algorithm fetches necessary information based on the query to provide relevant content for the LLM to process.

C. Language Learning Model Integration and Prompt Formulation

The LLM processes the information retrieved by the search algorithm, generating unstructured responses. These responses are captured. This step represents the system's capability to utilize advanced natural language processing techniques to synthesize and compile information. The prompts are tailored to enforce proper syntactical structure.

$$\begin{aligned}&\textit{Prompt } P_1 \textit{ is formulated to instruct LLM for}\\&\textit{augementation of all } n \textit{ retrieved chunks of type } S_1\\&\textit{consider augment method } A \textit{ such that from } (2),\end{aligned}$$

$$A([R(Q_1), R(Q_2), .. R(Q_N)]) = A\begin{bmatrix} S_{1-D_1} & \cdots & S_{1-D_N} \\ \vdots & \ddots & \vdots \\ S_{N-D_1} & \cdots & S_{N-D_N} \end{bmatrix} \quad (3)$$

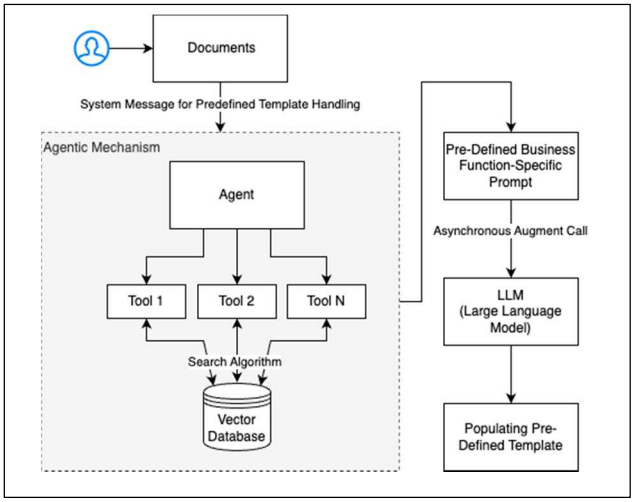

Fig. 1. Overview of the Agentic Mechanism for Document Generation

D. Document Synthesis

As a final stage, the augmented text is concatenated and applied with content formatting to meet the desired standards of pre-defined business template. This step transforms the LLM responses into a professionally formatted document.

$$\begin{aligned}&\textit{Now from } (3)\\&A([R(Q_1), R(Q_2), .. R(Q_N)]) = [S_1', S_2', .. S_N']\\&\textit{and from } (1)\\&f([S_1', S_2', .. S_N']) = D'\end{aligned}$$

In Fig 1. the mechanism architecture outlining the flow of data and augmentation. This completes the mechanism of the document synthesis experiment.

V. Observations

In this study, three search algorithms—xKNN, HNSW, and FullText—based on their latency, cost, and accuracy metrics were evaluated. The underlying LLM model used were Azure OpenAI GPT 4 Turbo and Azure OpenAI GPT4o[10]. Accuracy can be interpreted as the effectiveness of each algorithm in correctly identifying similar documents based on the proposed structural similarity measures [11][12]. To generate new documents, two base documents were utilized to assess the performance of the algorithms in synthesizing relevant content. This approach helped the evaluation of each algorithm's effectiveness in extracting and integrating information from existing documents.

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Table I. Performance metrics of different search algorithms

Algorithm	Metrics					
	Latency-GPT 4 Turbo (s)	*Latency - GPT 4o(s)*	*Cost - GPT 4 Turbo ($)*	*Cost - GPT 4o ($)*	*Accuracy (GPT 4 Turbo)*	*Accuracy (GPT 4o)*
xKNN	198	195	2.7	1.36	80%	78%
HNSW	208	206	2.8	1.43	93%	91%
FullText	199	200	2	1.01	69%	70%

In Table 1 The experiments were conducted for 2 input documents. Retrieval mechanism were benchmarked. In summary, HNSW is the most accurate algorithm among the three, while FullText is the most cost-effective option, highlighting a trade-off between performance and cost.

Table II. Performance metrics using HNSW search algorithms for number of input documents

No. of Base Documents	Metrics					
	Latency-GPT 4 Turbo (s)	*Latency - GPT 4o(s)*	*Cost - GPT 4 Turbo ($)*	*Cost - GPT 4o ($)*	*Accuracy - GPT 4 Turbo*	*Accuracy - GPT 4o*
2	208	206	2.8	1.43	93%	91%
3	252	240	2.6	1.36	89%	85%
4	303	298	2.2	1.22	90%	82%

In Table 2 described are the experiment results for multiple document inputs on increasing the document corpus the documents accuracy tested against the original documents decreases. A plausible inference can be made that on increasing input context the LLM models tend to hallucinate[13] resulting in erroneous behaviour.

VI. Conclusion

This study shows that using pre-existing Standard Operating Procedures (SOPs) from a business knowledge corpus can be a viable method for developing baseline SOPs. The goal of our experimental investigation was to create a generalized approach that could take in SOPs across different formats and output them comprehensively and specific to business functions. SOPs were generated in a scalable and economical manner by utilizing an Integrated Toolset design pattern in conjunction with Retrieval Augmented Generations.

Based on latency, cost, and accuracy measures, evaluated were three search algorithms: Full-Text search, Exhaustive K-Nearest Neighbors (*x*-KNN), and Hierarchical Navigable Small Worlds (HNSW). The results indicate HNSW is suitable for such generalised applications.

Augmenting using the Azure OpenAI GPT-4 Turbo and Azure OpenAI GPT4o models improved the generation process's efficacy by creating structurally related documents with decent accuracy. The cycle time, operating costs, and effort associated with creating SOPs have been shown to be significantly decreased by the structured approach to search algorithm selection and the integration of latent process steps from pre-existing SOPs. This study represents a significant advancement in digital operations across a range of business tasks by offering a strong framework for the production-grade operationalization of autonomous SOP generation systems and Natural Language Processing.

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